Infosys System Engineer: Hard-Level Mock Test — Paper 6

Total questions: 60 | **Duration:** 120 minutes | **Sectional cut-offs apply**

Section	Questions	Time
1. Reasoning Ability	Q1–15	25 min
2. Technical Ability (Quant)	Q16–25	35 min
3. Verbal Ability	Q26–45	20 min
4. Pseudo Code	Q46–50	10 min
5. Numeric Puzzle Solving	Q51–54	10 min
6. English Grammar	Q55–59	10 min
7. English Writing	Q60	10 min

SECTION 1: Reasoning Ability (15 Q, 25 min)

Q1. Statements: Some bugs are features. All features are releases. No release is stable. All patches are stable.

Conclusions: I. Some bugs are not stable. II. No patch is a feature. III. Some releases are bugs. IV. Some stable things are bugs.

(a) I and II only (b) II and III only (c) I, II and III only (d) All follow

Q2. *Data Sufficiency.* Is the product a × b positive?

I. a + b > 0.

II. a − b > 0.

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

Q3. *Data Sufficiency.* Is the positive integer n a perfect square?

I. n has exactly 3 positive divisors.

II. n is divisible by 9.

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

Q4. In a code, the 1st letter of a word is shifted 1 place forward, the 2nd letter 2 places forward, the 3rd letter 3 places forward, and so on. Using this rule, CODE is coded as DQGI. What is the code for LOGIC?

(a) MQJMH (b) MPJLH (c) MQJNH (d) MQIMH

Q5. In a code, the value of a word = (sum of the alphabetical positions of its letters × number of letters) − (number of vowels in the word). Given that CAB has the value 17, what is the value of DIGIT?

(a) 243 (b) 245 (c) 241 (d) 247

Q6. A solid cube of side 7 units is painted on all faces and cut into 1-unit cubes. How many small cubes have at most 1 painted face?

(a) 275 (b) 125 (c) 150 (d) 335

Q7. How many rectangles that are NOT squares can be found in a 4 × 5 grid of unit squares?

(a) 110 (b) 150 (c) 100 (d) 120

Q8. If P # Q means P is the mother of Q, P $ Q means P is the brother of Q, and P @ Q means P is the wife of Q, then in the expression A $ B # C @ D, how is A related to D?

(a) Wife's maternal uncle (uncle-in-law) (b) Brother-in-law (c) Father-in-law (d) Maternal uncle

Q9. A is 20 km due North of B. C is 15 km due East of B. What is the bearing of A from C (measured clockwise from North, to the nearest degree)?

(a) 323° (b) 307° (c) 037° (d) 217°

Q10. Five people A, B, C, D and E stand in a queue. A is ahead of B but behind C. D stands immediately behind B. E is ahead of C. Who stands third from the front?

(a) A (b) B (c) C (d) E

Q11. Find the next term: 3, 5, 11, 29, 83, ?

(a) 245 (b) 247 (c) 239 (d) 251

Q12. Find the next term: 1, 4, 27, 16, 125, 36, ?

(a) 343 (b) 49 (c) 216 (d) 64

Q13. Find the next group: ACF, GIL, MOR, ?

(a) SUX (b) SVY (c) TVY (d) SUW

Q14. A square sheet of paper is folded in half three times (each fold halves the previous folded shape). A single hole is then punched through all layers. When the sheet is fully unfolded, how many holes does it have?

(a) 8 (b) 4 (c) 6 (d) 16

Q15. *Data Sufficiency.* How is P related to Q?

I. P is the daughter of R.

II. R's brother is Q's father.

(a) Statement I alone is sufficient, but II alone is not

(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is

(d) Even both statements together are not sufficient

SECTION 2: Technical Ability, Quantitative (10 Q, 35 min)

Q16. In how many ways can 8 people be seated around a circular table so that two particular people are NOT seated next to each other?

(a) 3600 (b) 5040 (c) 1440 (d) 4320

Q17. How many 4-digit numbers (1000–9999) with all distinct digits are divisible by 5?

(a) 952 (b) 1008 (c) 896 (d) 504

Q18. Three fair dice are rolled. Given that at least one die shows a 6, what is the probability that the sum of the three dice is 12?

(a) 15/91 (b) 25/216 (c) 1/8 (d) 10/91

Q19. A disease affects 1% of a population. A test detects the disease in 95% of those who have it, but also gives a false positive for 10% of those who do not. If a randomly chosen person tests positive, what is the probability that they actually have the disease?

(a) 19/217 (b) 19/100 (c) 95/1000 (d) 1/11

Q20. A merchant sells two articles at ₹1,200 each. On one he gains 25%, on the other he loses 20%. What is his overall result (approximately)?

(a) 2.4% loss (b) 2.4% profit (c) No gain, no loss (d) 5% loss

Q21. A is twice as efficient as B, and B is three times as efficient as C. Working together, all three complete a job in 9 days. In how many days can C alone complete it?

(a) 90 days (b) 81 days (c) 60 days (d) 72 days

Q22. A 180 m long train passes a man walking at 6 km/h in the same direction in 12 seconds. How long will this train take to completely cross a 420 m long platform?

(a) 36 s (b) 30 s (c) 40 s (d) 42 s

Q23. A man spends 40% of his salary on food, 25% of the remainder on rent, and 20% of what is then left on travel. He is left with ₹3,600. What is his salary?

(a) ₹10,000 (b) ₹12,000 (c) ₹9,000 (d) ₹11,000

Q24. A loan is repaid in two equal annual installments of ₹2,420 each, with interest at 10% per annum compounded annually (first installment due at the end of year 1). What was the loan amount?

(a) ₹4,200 (b) ₹4,400 (c) ₹4,000 (d) ₹4,840

Q25. In what ratio must milk-water mixtures of strength 3:2 and 5:1 (milk:water) be mixed to obtain a mixture of strength 2:1?

(a) 5:2 (b) 2:5 (c) 3:2 (d) 1:3

SECTION 3: Verbal Ability (20 Q, 20 min)

Read the passage and answer Q26–30.

> *When a measure becomes a target, it ceases to be a good measure — a maxim known as Goodhart's law that engineering organisations rediscover with remarkable regularity. A team told that its performance will be judged by the number of closed tickets quickly learns to split large tasks into many trivial ones; a group rewarded for code coverage writes tests that execute lines without asserting anything meaningful. In neither case is the behaviour necessarily cynical. People respond rationally to the incentives they perceive, and the metric, once elevated to a goal, simply redirects that rationality away from the outcome it was intended to track. The remedy is not to abandon measurement, which would leave managers navigating by anecdote, but to treat any single metric as a symptom rather than a verdict, to pair it with a counter-metric that penalises its obvious abuse, and to revisit the pairing as teams learn how to game it. Even then, the arrangement is only ever provisionally sound, since every safeguard becomes, in time, another target.*

Q26. The passage's main point is that:

(a) Engineering metrics should be abolished

(b) Metrics that become targets get gamed, so they should be paired with counter-metrics and revisited, not dropped

(c) Developers deliberately deceive their managers

(d) Code coverage is the best available metric

Q27. The author's view of the gaming behaviour described is that it is:

(a) Usually a sign of dishonesty

(b) Not necessarily cynical, but a rational response to perceived incentives

(c) Impossible to prevent

(d) Limited to junior engineers

Q28. Treating a metric "as a symptom rather than a verdict" most nearly means:

(a) Ignoring the metric entirely

(b) Using the metric as a signal that prompts investigation rather than a final judgement of performance

(c) Replacing the metric every quarter

(d) Publishing the metric to all staff

Q29. According to the passage, why would abandoning measurement be undesirable?

(a) It would reduce team morale

(b) It would make teams ignore tickets

(c) Managers would be left relying on anecdote

(d) It would violate Goodhart's law

Q30. The final sentence implies that:

(a) Even a counter-metric can eventually be gamed, so the arrangement needs periodic review

(b) Counter-metrics are worse than the original metrics

(c) Safeguards should never be introduced

(d) Teams eventually stop responding to incentives

Sentence Correction (Q31–35).

Q31. Neither the project manager nor the developers **was informed** about the change.

(a) No improvement (b) were informed (c) is informed (d) has been informed

Q32. Walking into the server room, the temperature seemed unusually high to the technician.

(a) No improvement (b) the technician found the temperature unusually high (c) the temperature seemed to the technician unusually high (d) it was unusually high to the technician

Q33. The new framework is faster, more secure, **and it is easy to maintain**.

(a) No improvement (b) and easier to maintain (c) and also is easy for maintenance (d) and maintaining it is easy

Q34. By the time the patch was released, the attackers **already exploited** the vulnerability.

(a) No improvement (b) had already exploited (c) have already exploited (d) were already exploiting

Q35. The reason why the build failed is because of a missing dependency.

(a) No improvement (b) The reason the build failed was because a dependency was missing (c) The build failed because a dependency was missing (d) The reason why the build failed was because of a missing dependency

Para-jumbles (Q36–38).

Q36.

P: Consequently, many teams now treat flaky tests as defects in their own right.

Q: A test that passes and fails intermittently without any code change erodes trust in the whole suite.

R: Developers begin to ignore failures, assuming they are noise.

S: Real regressions then slip through unnoticed.

(a) QRSP (b) QSRP (c) RQSP (d) QRPS

Q37.

P: Releasing to every user at once maximises speed but also the blast radius of any defect.

Q: Teams face a genuine tension between releasing quickly and releasing safely.

R: Releasing to a handful of users first limits risk but delays feedback.

S: A staged rollout, exposing a small and growing share of users, is the usual compromise.

(a) QPRS (b) QRPS (c) PQRS (d) QPSR

Q38.

P: Its adoption, however, was slow outside the research community.

Q: Public-key cryptography solved the problem of exchanging secret keys over an insecure channel.

R: Usable libraries and standards eventually removed this barrier.

S: Early implementations were notoriously difficult to use correctly.

(a) QSPR (b) QPSR (c) PQSR (d) QPRS

Vocabulary, Synonyms and Antonyms (Q39–45).

Q39. The documentation was admirably **pellucid**, leaving no step open to misreading. "Pellucid" means:

(a) lengthy (b) transparently clear (c) outdated (d) technical

Q40. The client's **mercurial** moods made planning difficult. "Mercurial" means:

(a) steady (b) unpredictably changeable (c) hostile (d) cheerful

Q41. Fill in the blank: The startup's growth was _____; revenue tripled in a single quarter.

(a) meteoric (b) gradual (c) stagnant (d) negligible

Q42. Synonym of **EQUIVOCATE**:

(a) speak ambiguously to avoid commitment (b) confess openly (c) argue loudly (d) agree readily

Q43. Synonym of **PERNICIOUS**:

(a) harmless (b) subtly or gravely harmful (c) trivial (d) beneficial

Q44. Antonym of **BENEVOLENT**:

(a) kind (b) malevolent (c) generous (d) charitable

Q45. Antonym of **EXACERBATE**:

(a) worsen (b) aggravate (c) alleviate (d) intensify

SECTION 4: Pseudo Code (5 Q, 10 min)

Q46. What is printed? (`/` is integer division, `%` is modulo)

```
int f(int n) {
    if (n <= 1) return n;
    if (n % 2 == 0) return f(n / 2);
    return f(n - 1) + f(n + 1);
}
print(f(11))
```

(a) 5 (b) 4 (c) 6 (d) 3

Q47. What is printed? (^=XOR, >>=right shift, &=AND)

```
n = 13
g = n ^ (n >> 1)
print(g + (g & 5))
```

(a) 12 (b) 11 (c) 16 (d) 14

Q48. What is printed?

```
s = 0
for i = 1 to 5
    for j = 1 to 5
        if ((i + j) % 4 == 0) continue
        if (j > i + 1) break
        s = s + i * j
    end for
end for
print(s)
```

(a) 142 (b) 138 (c) 146 (d) 150

Q49. The following sorts a stack using a temporary stack (the top of a stack is its last element). How many times is an element moved from `tmp` back to `st`?

```
st = [3, 1, 4, 2]; tmp = []; moves = 0
while (st is not empty)
    t = pop(st)
    while (tmp is not empty and top(tmp) > t)
        push(st, pop(tmp))
        moves = moves + 1
    end while
    push(tmp, t)
end while
print(moves)
```

(a) 3 (b) 2 (c) 4 (d) 5

Q50. What is printed? (dynamic programming for the minimum number of coins; INF is a very large number)

```
coins = [1, 3, 4]
dp[0] = 0; dp[1..6] = INF
for i = 1 to 6
    for each c in coins
        if (i >= c)
            dp[i] = min(dp[i], dp[i - c] + 1)
        end if
    end for
end for
print(dp[6])
```

(a) 2 (b) 3 (c) 4 (d) 6

SECTION 5: Numeric Puzzle Solving (4 Q, 10 min)

Q51. *Matchstick.* A 2 × 3 grid of unit squares (2 rows, 3 columns) is built from 17 matchsticks and contains 8 squares in total (six 1×1 and two 2×2). What is the minimum number of sticks that must be removed so that no complete square of any size remains?

(a) 3 (b) 4 (c) 5 (d) 6

Q52. *Circular wheel.* Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.

Wheel 1: T = 4, R = 1, B = 3, L = 2, Centre = 40

Wheel 2: T = 6, R = 2, B = 2, L = 3, Centre = 39

Wheel 3: T = 5, R = 3, B = 4, L = 1, Centre = ?

(a) 65 (b) 61 (c) 69 (d) 57

Q53. *Matrix logic.* Find the missing number.

5	7	18
6	11	25
8	9	?

(a) 55 (b) 73 (c) 64 (d) 47

Q54. *Pattern.* Find the next term: 2, 10, 30, 68, 130, ?

(a) 222 (b) 216 (c) 210 (d) 228

SECTION 6: English Grammar (5 Q, 10 min)

Q55. The quality of the algorithms, along with the efficiency of the data structures, _____ determined the system's throughput.

(a) have (b) has (c) were (d) are

Q56. No sooner _____ the alert been raised than the on-call engineer responded.

(a) had (b) has (c) did (d) was

Q57. He is oblivious _____ the risks, though he was warned _____ them repeatedly, and he insists _____ proceeding.

(a) of, about, in (b) to, of, on (c) to, for, in (d) with, of, on

Q58. *Error spotting.* "Despite of the repeated warnings (A) / the team continued to deploy (B) / changes directly to production, (C) / which was clearly risky. (D)"

(a) A (b) B (c) C (d) D

Q59. _____ European Union has adopted _____ unique approach to regulating _____ artificial intelligence, and _____ honest debate about its merits continues.

(a) The, a, -, an (b) A, a, the, an (c) The, an, -, a (d) The, a, the, a

SECTION 7: English Writing (1 Q, 10 min)

Q60. *Write an email of 150–180 words.*

While reviewing a client's production application, you discover a security vulnerability that could expose customer data if exploited. Write an email to the client's CTO that (i) informs them of the issue clearly but without causing panic or assigning blame, (ii) states the immediate mitigation you have already applied, (iii) outlines the permanent fix you propose, and (iv) requests an urgent meeting.

Evaluation rubric (10 marks): Subject line, format and tone (2) | Clear, calm explanation of the issue and risk (2) | Mitigation and permanent fix described (3) | Clear call to action (1) | Grammar, vocabulary and word limit (2)

ANSWER KEY WITH EXPLANATIONS

Section 1: Reasoning

1. **(c)** Some bugs are features, features are releases, and no release is stable — so those bugs are not stable (I). Patches⊆stable and features∩stable=∅ (features are releases), so no patch is a feature (II). Some bugs are releases, which converts to "some releases are bugs" (III). IV does not follow: the only bugs we know are unstable; nothing says any bug is stable.

2. **(d)** a=3, b=1 satisfies both statements and gives a positive product; a=3, b=−1 also satisfies both (a+b=2>0, a−b=4>0) but gives a negative product. Even together they are not sufficient.

3. **(a)** I: a number with exactly 3 divisors must be the square of a prime (p^2 has divisors 1, p, p^2), so n is a perfect square — sufficient. II: n=18 is divisible by 9 but is not a perfect square, while n=9 is — insufficient.

4. **(a)** L+1=M, O+2=Q, G+3=J, I+4=M, C+5=H, giving **MQJMH** (verified: the same rule turns CODE into DQGI).

5. **(a)** DIGIT: positions 4+9+7+9+20 = 49; 49×5 = 245; vowels (I, I) = 2; 245−2 = **243** (check: CAB = 6×3−1 = 17 ✓).

6. **(a)** 0 painted faces (interior) = 5^3 = 125; exactly 1 painted face = 6×5^2 = 150; total = **275**.

7. **(a)** Total rectangles = C(5,2)×C(6,2) = 10×15 = 150. Squares = 20+12+6+2 = 40. Non-square rectangles = 150−40 = **110**.

8. **(a)** A is the brother of B, and B is the mother of C, so A is C's maternal uncle. C is the wife of D, so A is D's wife's maternal uncle — an **uncle-in-law**.

9. **(a)** Taking East and North as axes, A is at (0,20), B at (0,0), C at (15,0). The vector from C to A is (−15, 20): 36.87° west of north, giving a bearing of 360°−36.87° ≈ **323°**.

10. **(a)** E is ahead of C, C ahead of A, A ahead of B, and D immediately behind B: E, C, A, B, D. Third from the front is **A**.

11. **(a)** Differences are 2, 6, 18, 54 (each ×3); the next difference is 162. 83+162 = **245**.

12. **(a)** Odd positions are cubes (1^3, 3^3, 5^3) and even positions are squares (2^2, 4^2, 6^2). The 7th term is 7^3 = **343**.

13. **(a)** Each group has three letters with gaps +2, +3 (A→C→F, G→I→L, M→O→R), and each group starts 6 letters after the previous one. The next group is S, U, X = **SUX**.

14. **(a)** Each fold doubles the layers: 2^3 = 8 layers, so one punch makes **8** holes.

15. **(c)** I alone gives only P's parent; II alone links R and Q's father without involving P. Together: P is the daughter of R, and R is the sibling of Q's father, so P and Q are **cousins**.

Section 2: Quant

16. **(a)** Total circular arrangements = 7! = 5040. Arrangements with the two people adjacent = 2×6! = 1440. Not adjacent = 5040−1440 = **3600**.

17. **(a)** Last digit 0: 9×8×7 = 504. Last digit 5: the first digit has 8 choices (not 0 or 5), then 8 and 7 for the middle: 8×8×7 = 448. Total = 504+448 = **952**.

18. **(a)** Outcomes with at least one 6: $6^3-5^3 = 91$. Of these, the sum is 12 in 15 outcomes (permutations of {6,5,1}: 6; {6,4,2}: 6; {6,3,3}: 3). Probability = **15/91**.

19. **(a)** P(disease and positive) = 0.01×0.95 = 0.0095; P(no disease and positive) = 0.99×0.10 = 0.099. P(disease | positive) = 0.0095/0.1085 = **19/217** (≈ 8.8%).

20. **(a)** $CP_1 = 1200/1.25 = 960$; $CP_2 = 1200/0.80 = 1500$. Total CP = 2460, total SP = 2400, loss = 60. Loss % = 60/2460 ≈ **2.4% loss**.

21. **(a)** Let C = 1 unit/day, B = 3, A = 6, so together 10 units/day. In 9 days the work is 90 units, so C alone needs 90/1 = **90 days**.

22. **(a)** Relative speed = 180/12 = 15 m/s = 54 km/h; the train's speed = 54+6 = 60 km/h = 50/3 m/s. Time to cross the platform = (180+420)/(50/3) = **36 s**.

23. **(a)** After food: 60%. After rent (25% of 60% = 15%): 45%. After travel (20% of 45% = 9%): 36%. 36% of salary = 3600, so the salary = **₹10,000**.

24. **(a)** Loan = $2420/1.1 + 2420/1.1^2 = 2200 + 2000$ = **₹4,200**.

25. **(a)** Milk fractions: 3/5, 5/6, target 2/3. By alligation, the amounts are in the ratio (5/6−2/3) : (2/3−3/5) = 1/6 : 1/15 = **5:2**.

Section 3: Verbal

26. **(b)** 27. **(b)** 28. **(b)** 29. **(c)** 30. **(a)**

31. **(b)** With "neither…nor", the verb agrees with the nearer subject (developers → were).

32. **(b)** The opening participle must modify a person; "the technician found the temperature…" fixes the dangling modifier.

33. **(b)** The adjectives must stay parallel: faster, more secure, and easier to maintain.

34. **(b)** An action completed before another past action needs the past perfect: "had already exploited".

35. **(c)** "The reason… is because" is redundant; "The build failed because a dependency was missing" is the clean form.

36. **(a)** Q→R→S→P. 37. **(a)** Q→P→R→S. 38. **(b)** Q→P→S→R.

39. **(b)** 40. **(b)** 41. **(a)** 42. **(a)** 43. **(b)** 44. **(b)** 45. **(c)**

Section 4: Pseudo Code

46. **(a) 5.** Building up: f(1)=1, f(2)=f(1)=1, f(4)=f(2)=1, f(3)=f(2)+f(4)=2, f(6)=f(3)=2, f(5)=f(4)+f(6)=3, f(10)=f(5)=3, f(12)=f(6)=2. So f(11)=f(10)+f(12)=3+2=**5** (confirmed by code).

47. **(a) 12.** 13 = 1101_2; 13>>1 = 6 = 0110_2; g = 1101 ^ 0110 = 1011_2 = 11. g & 5 = 1011 & 0101 = 0001 = 1. g + 1 = **12**.

48. **(a) 142.** Tracing the loops with the `continue` (when i+j is a multiple of 4) and `break` (when j > i+1) gives a total of **142** (confirmed by code).

49. **(a) 3.** Tracing: pop 2 → tmp [2]; pop 4 → tmp [2,4]; pop 1 → move 4 and 2 back (2 moves) → tmp [1]; pop 2 → tmp [1,2]; pop 4 → tmp [1,2,4]; pop 3 → move 4 back (1 move), tmp [1,2,3]; pop 4 → tmp [1,2,3,4]. Total moves = **3**.

50. **(a) 2.** dp = [0, 1, 2, 1, 1, 2, 2], so dp[6] = **2** (3+3). A greedy approach (4+1+1) would give 3, which is why the DP matters.

Section 5: Numeric Puzzles

51. **(b) 4.** Exhaustive computer search over all 17 sticks and all 8 squares shows that no set of 3 sticks breaks every square, while a set of 4 does. Minimum = **4**.

52. **(a) 65.** Rule: Centre = $(T+B)^2 - (R+L)^2$. Wheel 1: $7^2-3^2 = 40$ ✓. Wheel 2: $8^2-5^2 = 39$ ✓. Wheel 3: $9^2-4^2 = 81-16 =$ **65**.

53. **(a) 55.** Rule: third number = $(\text{first})^2 - \text{second}$. Row 1: 25−7 = 18 ✓. Row 2: 36−11 = 25 ✓. Row 3: 64−9 = **55**.

54. **(a) 222.** Each term is n^3+n: 2, 10, 30, 68, 130; for n=6: 216+6 = **222**.

Section 6: Grammar

55. **(b)** The true subject is "the quality" (singular); "along with…" does not make it plural → has.

56. **(a)** "No sooner" takes the past perfect with inversion: "No sooner had the alert been raised than…".

57. **(b)** oblivious **to**, warned **of**, insists **on**.

58. **(a)** "Despite of" is incorrect; it should be "Despite" or "In spite of".

59. **(a)** The European Union (proper institution), a unique (starts with /j/ sound), artificial intelligence (general concept, no article), an honest (silent h).

Section 7: Writing

Use the rubric above. A strong email has a calm, factual subject line (e.g., "Security Finding in Production Application — Action Needed"), states the vulnerability and potential impact neutrally (no blame), describes the immediate mitigation already applied (for example, restricting access or applying a temporary rule), proposes the permanent fix with a rough timeline, and closes by asking for an urgent meeting at specific times.

Scoring guide

Award +1 per MCQ (Q1–59) and up to 10 for Q60. This is the hardest paper in the series. The matchstick answer (Q51), the dice, loan and probability answers, and the pseudo-code traces (Q46–50) were verified by running the logic in code, so trust the key even where the working looks dense.